

A Packing of 9 Four-Spaces in $\mathbb{R}^{12}$

Limin Ge*

October 9, 2026

Abstract

We give 9 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.999732. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.962710, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 9 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.999732 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 9$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.962710, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 9/8 = 3.000000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N9.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
149986620243152302343239578864097032974679969401755708599014521013345502
351038599191770326593484561762263283690930566020132636495102932307792323
2617949151653279125866760269122567889681059720383
```

denominator:

```
499999999998599291785704362331812300190691913910579502386657632351251311
100283952450896075792796798703218696537772598547128251034054898380432378
420350242481818858428695512006290890581407754035
```

B Coordinates

9 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N9.txt`.

```
438173219560 307187227102 361424826440 8450075254 -91110123015 98869805254 433124755104 -2573457355467983079865 321157472736 407851431413 360123754499
125889588648 61537882594 -546836082790 38927416217611409367994 -343087398100 -103878684762 1291138544390379034484 -355403171967 -67888922169 77079895700
460812719078 6019233691 -238229653643 -498997540645-123114599801 296843430973 -239581201063 7138540571766204455367 -302861261567 411758644756 193614337351
-375618732651 315994132976 -134924297200 -86130689520801766737 137564323222 35447836967 374193347719 -433136727871 3188659881 -348295426918 -90998196367
289486124653 -524395765867 141710799180 3208567836640497717896 110859578188 53414816403 -217750933369-396752169822 314384377368 23419424710 -245826700021
-153102387017 -338030274417 -146273825845 2644465795319805447860 -1500796027 -56872682896 -32947054169-278608740774 -156154670623 -51433000752 683405231162
417011302494 -213320570171 110891697700 94055841279-175064338381 -269488018390 102106301298 137669049085499631081 409567277412 163005917569 -68023751240
-300943251892 -189748783459 353325183756 -42148575322643100053 370175583537 664717179816 10866843327584993077981 48158007658 -245050002269 -32437804242
68121433260 -257425390757 -464722962061 -42459595215228001304545 149661839170 265117613877 51434223307270299969332 -41917920464 -139081104766 -291042298864
152600439284 360618837977 -248689086351 47708948712529901829726 696488782218 -436709209050 5883508262766439491496 -53184383487 61793207748 -193364392420
23656490538 -86880587749 99059755072 15157870365 444938938849 -184997781529 -131564884614 5990936637920003626903 549224158411 -435093768116 373467020290
-196273709193 -356837366961 -170717001213 1863455274459520124921 42550101551 79262806038 256516971901 -222549420338 281588562530 485706311594 -154520051211
-457424525146 -89753816439 -439615700874 -337200783465313207233 369292974812 -33393714002 -1846391012583932862203 -260063434659 59228348308 170124939482
-336543460873 -173451032816 -217445640857 761182927133073376474 216341386182 -200824006761 6729918673687623962064 94237707598 -559321997197 -49787424753
-19173339624 -419488010939 -303486397453 -8180807351555540912534 -41714601242 185856562843 18650397488435268082985 39931767732 203026943794 307953083008
-244031442258 183422595129 41970138024 -614205838849075125693 123669914329 544789029705 -2484144203820355158795 604742602075 38993399647 80066338438
435684147934 -52616016623 -149195078995 -1385802250888611737 -296572220804 201044237905 79383436984 -341313048065 -288871741958 21676456851 653871419992
-137815187665 16060095813 -479976623180 -226368026888134503180 -5324211482 -300696458029 2403441508393111503875 -189274924054 247731897646 -527169995516
-167706952337 430224002484 -171417109284 5804259960224983241511 -296104192074 602654511794 30208014478135585859 426069194211 119284955206 14087700759
122081701738 16481936222 -279551612109 -2280846777460717976389 383325612215 321732879367 253730609902-322934022392 -156821936386 -7353442857 -350914310683
549709307034 -165706166504 -306318232784 4984753530409787757079 241960686749 -46678603998 207268346019827928105 -288603861946 -634347932163 -25098802489
-50062771287 112814323677 146904983871 -3243159309081737286122 -369073141631 -82135552807 16705358128317814906891 -376752486145 349846629981 -3382885302
-205665656935 -705118134884 169609414037 13602671103571619920805 234912859456 75126380381 33881700770396461494313 -74451164141 182822721375 -106007521539
-136945641588 -30701466125 401548248603 -16141478309911755329 -473746300673 -122906726758 319487907685348984759 737988467 -81070552809 -163274781234
63535184768 129954321869 431348744716 205253075934 82302455806 555944667951 28213888379 -157560734767 -388044604956 71637786131 255205176062 -278698469436
-100844666694 -390427740707 178229010339 -55203388401287248709 -2497267811175 -88391681244 -20821426296792875110 576264210704 -48451959487 -373993100866
-177354924619 615150528072 -390944935739 13941144129034961059800 -115156192836 -98739846452 -6058782019295106019 364327603899 -528861251318 273954307771
-382202129029 294873880949 596780737781 999205664026944310308 40513833513 12798439216 -83680349398 -98222521190 448770749527 267515202589 233427516086
-266238960122 83140023405 282217205957 40146656750 109967863715 37013314645 433211208457 112454133185 260296213165 160794108795 -150967827129 -626104375779
53788685939 139085987722 75460169826 454346499952 -258551184673 -286415518445 411443486018 35175666038255669513355 205994747946 110536780259 -45094109921
39971785781 410202982085 -188911477086 -2882145904352780627386 -278798762197 -83082421332 4344662022158975791699 241614554227 468790927351 215509138390
408613927196 -8463956670 -88564168483 474572619903 372644454489 -102341174291 185933869033 -2961598797441077824658 -4862741974 -186288497342 229268493972
10676330814 252951877606 189689590942 -595314945174569992822 523299807632 -244332324387 2959793057923132332114 -213187896792 342615999646 54219073779
649725791231 89543275632 282163955183 115476064109 -112007434509 -29821156967 -521627423605 6594977561076536779201 189365393468 344542656302 -113347854431
-145034696659 76010228069 -37162366301 7275003458 -495458264221 -338200977775 -148319364450 -1367954208883464387 -334096057812 124682238922 -302913311849
-333682231047 -304863526431 -165335603760 688752568683394189031 230090307537 478804403165 -1849710615281485049480 95076129288 200362398210 228745793158
```